LIST OF OPEN ELECTIVESOFFERED BY VARIOUS DEPARTMENTS TO OTHER DEPARTMENTSIN IV YEAR I SEMESTER OE-IV

Offered by	Course Code	Course Name	Offered to
ARTIFICIAL INTELLIGENCE & DATA SCIENCE	**B23ADOE06**	Principles of Internet of Things	**CE, ECE, EEE &ME**
	B23QTOE01	Quantum Science and Technology	**All the Branches**
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING	**B23AMOE08**	IOT Based Smart Systems	**CE, ECE, EEE &ME**
	B23AMOE09	Computer Networks	
	B23AMOE10	Software Engineering	
	B23QTOE01	Quantum Science and Technology	**All the Branches**
CIVIL ENGINEERING	**B23CEOE07**	Alternative Energy Sources	**AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME**
	B23CEOE08	Water Supply Systems	
COMPUTER SCIENCE & BUSINESS SYSTEMS	**B23CBOE05**	Principles Of Operating Systems	**CE, ECE, EEE &ME**
	B23QTOE01	Quantum Science and Technology	**All the Branches**
COMPUTER SCIENCE & ENGINEERING	**B23CSOE05**	Principles of Software Engineering	**CE, ECE, EEE &ME**
	B23CSOE06	Computer Networks	**CE, ECE, EEE &ME**
	B23QTOE01	Quantum Science and Technology	**All the Branches**
CSE (Internet of Things and Cyber Security including Block Chain Technology)	**B23CIOE07**	Introduction to Cyber Security	**CE, ECE, EEE &ME**
	B23CIOE08	Introduction to Block Chain Technology (Prerequisites: CNS)	
	B23QTOE01	Quantum Science and Technology	**All the Branches**
ELECTRONICS & COMMUNICATION ENGINEERING	**B23ECOE06**	Principles of Cellular & Mobile Communications	**AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT & ME**
	B23ECOE07	Image Processing	
	B23ECOE08	Fundamentals of Satellite Communications	
ELECTRICAL & ELECTRONICS ENGINEERING	**B23EEOE07**	Introduction to Drone Technology	**AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT & ME**
	B23EEOE08	Energy Management and Audit	
INFORMATION TECHNOLOGY	**B23ITOE04**	Principles Of Operating Systems	**CE, ECE, EEE &ME**
	B23QTOE01	Quantum Science and Technology	**All the Branches**
MECHANICAL ENGINEERING	**B23MEOE09**	Additive Manufacturing	**AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE & IT**
	B23MEOE10	Industrial Safety	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ADOE06	OE	3	-	-	3	30	70	3 Hrs.

PRINCIPLES OF INTERNET OF THINGS

(Offered by AIDS)

(Offered to CE, ECE, EEE &ME)

Course Objectives: This course is designed to:

1	Provide a clear understanding of IoT concepts, architecture, M2M communication, and basic communication technologies.
2	Explain web connectivity in IoT, including internet principles, IP addressing, and application layer protocols.
3	Introduce IoT connectivity and communication technologies including Bluetooth, Zigbee, Wi-Fi, LoRa, NB-IoT, RFID, NFC, and various IoT communication protocols.
4	Provide knowledge of IoT sensing, actuation, data processing, and processing topologies in IoT systems.
5	Enable understanding of IoT paradigms, challenges, and real-world applications in domains such as agriculture, healthcare, and vehicular systems.

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Understand** IoT concepts, architecture, M2M communication, and basic communication technologies.	K2
2.	**Explain** web connectivity, internet protocols, and communication mechanisms used in IoT systems.	K2
3.	**Apply** IoT connectivity and communication technologies, including Bluetooth, Zigbee, Wi-Fi, LoRa, NB-IoT, RFID, NFC, and IoT communication protocols in system design.	K3
4.	**Demonstrate** IoT sensing, actuation, and data processing techniques including processing topologies.	K2
5.	**Analyse** IoT paradigms, challenges, and real-world applications in agriculture, healthcare, and vehicular systems.	K3

SYLLABUS

UNIT-I (10Hrs)	**The Internet of Things**: Internet of Things, IoT Conceptual Framework, IoT Architectural view, Technology behind IoT, Sources of IoT, M2M Communication, Examples OF IoT. **Design Principles for Connected Devices**: Introduction, Modified OSI Model for the IoT/M2M Systems, ITU-T Reference Model, ETSI M2M Domains and High-level Capabilities, Basic Communication Technologies- Wired and Wireless communication technologies.

UNIT-II (8 Hrs)	**Design Principles for the Web Connectivity**: Web Communication protocols for Connected Devices, Message Communication protocols for Connected Devices, Web Connectivity for connected Devices. **Internet Connectivity Principles**: IP Addressing in IoT, Media Access Control (MAC), Application Layer Protocols: HTTP, HTTPS, FTP, Telnet.
UNIT-III (10 Hrs)	**IoT Connectivity Technologies**: Introduction, Short-range communication: Bluetooth, Zigbee, Wi-Fi, Long-range communication: LoRa, NB-IoT, Identification technologies: RFID, NFC. **IoT Communication Technologies**: Introduction, Infrastructure Protocols, Discovery Protocols, Data Protocols, Identification Protocols, Device Management, Semantic Protocols.
UNIT-IV (10 Hrs)	**IoT Sensing and Actuation**: Introduction, Sensors, Sensor Characteristics, Sensorial Deviations, Sensing Types, Sensing Considerations, Actuators, Actuator Types, Actuator Characteristics. **IoT Processing Topologies and Types**: Data Format, Importance of Processing in IoT, Processing Topologies, IoT Device Design and Selection Considerations, Processing Offloading.
UNIT-V (8 Hrs)	**Paradigms, Challenges, and the Future**: Introduction, Evolution of New IoT Paradigms, Emerging Pillars of IoT, Challenges Associated with IoT. IoT Case Studies: Agricultural IoT, Vehicular IoT, Health care IoT.
Textbooks:	
1.	Internet of Things: Architecture and Design Principles, Rajkamal, McGraw Hill Higher.
2.	Education Introduction to IoT, Sudip Misra, Anandarup Mukhaerjee, Arjit Roy, Cambridge University Press, 2021.
Reference Books:	
1.	Fog and Edge Computing: Principles and Paradigms, Rajkumar Buyya (Editor), Satish Narayana Srirama (Editor) , ISBN: 978-1-119-52498-4, January 2019
2.	Getting Started with the Internet of Things, CunoPfister , Oreilly

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AMOE08	OE	3	--	--	3	30	70	3 Hrs.

IOT BASED SMART SYSTEMS

(Offered by AIML)

(Offered to CE, ECE, EEE &ME)

Course Objectives:

1.	To study about Internet of Things technologies and its role in real time applications.
2.	To introduce the infrastructure required for IoT
3.	To familiarize the accessories and communication techniques for IoT.
4.	To provide insight about the embedded processor and sensors required for IoT.
5.	To familiarize the different platforms and Attributes for IoT.

Course Outcomes: At the end of the course, Students will be able to

S.No	Outcome	Knowledge Level
1.	Apply IoT concepts and present developments to real-world applications.	K3
2.	Apply different IoT platforms and infrastructures for suitable IoT solutions.	K3
3.	Explain different protocols and communication technologies used in IoT.	K2
4.	Apply big data analytics and IoT programming techniques in IoT environments.	K3
5.	Implement IoT solutions for smart applications	K3

SYLLABUS

UNIT-I (10Hrs)	**INTRODUCTION TO INTERNET OF THINGS:** Overview, Hardware and software requirements for IOT, Sensor and actuators, Technology drivers, Business drivers, Typical IoT applications, Trends and implications.
UNIT-II (10 Hrs)	**IOT ARCHITECTURE:** IoT reference model and architecture -Node Structure – Sensing, Processing, Communication, Powering, Networking – Topologies, Layer/Stack architecture, IoT standards, Cloud computing for IoT, Bluetooth, Bluetooth Low Energy beacons.
UNIT-III (10 Hrs)	**PROTOCOLS AND WIRELESS TECHNOLOGIES FOR IOT PROTOCOLS:** NFC, SCADA and RFID, Zigbee MIPI, M-PHY, UniPro, SPMI, SPI, M-PCIe GSM, CDMA, LTE, GPRS, small cell. **Wireless technologies for IoT:** Wi-Fi (IEEE 802.11), Bluetooth/Bluetooth Smart, ZigBee Smart, UWB (IEEE 802.15.4), 6LoWPAN, Proprietary systems-Recent trends.
UNIT-IV (10 Hrs)	**IOT PROCESSORS Services/Attributes:** Big-Data Analytics for IOT, Dependability, Interoperability, Security, Maintainability. Embedded processors for IOT: Introduction to Python programming -Building IOT with RASPERRY PI and Arduino.

UNIT-V (10 Hrs)	**CASE STUDIES:** Industrial IoT, Home Automation, smart cities, Smart Grid, connected vehicles, electric vehicle charging, Environment, Agriculture, Productivity Applications, IOT Defense.

Textbooks:

1.	ArshdeepBahga and VijaiMadisetti : A Hands-on Approach “Internet of Things”,Universities Press 2015.
2.	Oliver Hersent , David Boswarthick and Omar Elloumi “ The Internet of Things”, Wiley,2016.
3.	Samuel Greengard, “ The Internet of Things”, The MIT press, 2015.

Reference Books:

1.	Adrian McEwen and Hakim Cassimally“Designing the Internet of Things “Wiley,2014.
2.	Jean- Philippe Vasseur, Adam Dunkels, “Interconnecting Smart Objects with IP: The Next Internet” Morgan Kuffmann Publishers, 2010.
3.	Adrian McEwen and Hakim Cassimally, “Designing the Internet of Things”, John Wiley and sons, 2014.
4.	Lingyang Song/DusitNiyato/ Zhu Han/ Ekram Hossain,” Wireless Device-to-Device Communications and Networks, CAMBRIDGE UNIVERSITY PRESS,2015.
5.	OvidiuVermesan and Peter Friess (Editors), “Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers Series in Communication, 2013.
6.	Vijay Madisetti ,ArshdeepBahga, “Internet of Things (A Hands on-Approach)”, 2014.

e-Resources

1.	https://onlinecourses.nptel.ac.in/noc26_cs37/preview
2.	https://www.eicta.iitk.ac.in/courses/internet-of-things-iot-2

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AMOE09	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

COMPUTER NETWORKS

(Offered by AIML)

(Offered to CE, ECE, EEE &ME)

Course Objectives: This course aims to equip students with the following:

1.	To understanding the principles of computer networks.
2.	To familiarize with Reference model OSI and TCP/IP
3.	To explore Datalink, Transport and Network layer protocols
4.	To study application layer applications

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	Illustrate the OSI reference model, TCP/IP, and Digital transmission techniques	**K2**
2.	Apply error detection and correction, flow control with respect to data link layer	**K3**
3.	Summarize MAC layer protocols and LAN technologies	**K2**
4.	Demonstrate various network layer services and Routing algorithms	**K3**
5.	Explain Transport layer and application layer protocols	**K2**

SYLLABUS

UNIT-I (10Hrs)	**Introduction:** Types of Computer Networks, Network Topologies Reference Models- The OSI Reference Model, The TCP/IP Reference Model, A Comparison of the OSI and TCP/IP Reference Models. **Physical Layer**: Introduction to physical layer, Guided Media- Twisted-pair cable, Coaxial cable and Fiber optic cable and unguided media
UNIT-II (10 Hrs)	**The Data Link Layer:** Data Link Layer Design Issues, Services Provided To the Network Layer, Error detecting and Error Correcting codes, Elementary Data Link Protocols, Sliding Window Protocols, HDLC. Multiple Access Protocols in Wired Lans, Ethernet, Fast Ethernet, Gigabit Ethernet
UNIT-III (10 Hrs)	**The Network Layer**: Network Layer Design Issues, Routing Algorithms, Congestion, Congestion control algorithms. The Network Layer in the Internet, The IP Version 4 Protocol, IP Addresses- Classful, CIDR, NAT, IP Version 6 Protocol, Transition from IPV4 to IPV6
UNIT-IV (10 Hrs)	**The Transport Layer:** The Transport Layer Services, Connection Establishment and Termination, Congestion Control, Sliding Window Protocol, Transport Layer Protocols: UDP, TCP and SCTP

UNIT-V (10 Hrs)	**The Application Layer:** Services And Protocols, The World Wide Web, HTTP, Domain Name Space, Remote Loging, Electronic Mail and File Transfer
Textbooks:	
1.	"Computer Networks", Andrew S Tanenbaum, David J Wetherall, 5th Edition, Pearson
2.	"Data Communications and Networking", Behrouz A Forouzan, 4th Edition, Tata McGraw Hill Education
Reference Books:	
1.	"Data and Computer Communication", William Stallings, Pearson
2.	"TCP/IP Protocol Suite", Behrouz Forouzan, McGraw Hill.
e-Resources	
1.	https://nptel.ac.in/courses/106105183/25
2.	http://www.nptelvideos.in/2012/11/computer-networks.html
3	https://www.youtube.com/playlist?list=PLBlnK6fEyqRiw-GZRqfnlVIBz9dxrqHJS

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AMOE10	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

SOFTWARE ENGINEERING

(Offered by AIML)

(Offered to CE, ECE, EEE &ME)

Course Objectives: This course aims to equip students with the following:

1.	Give exposure to phases of software development and common process models.
2.	Give exposure to a variety of software engineering practices such as requirements analysis and specification
3.	Give exposure to software planning strategies
4.	Give exposure to software design techniques
5.	Give exposure to various software testing methods and strategies

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	Demonstrate the key concepts of software engineering and software development process models in real-world scenarios.	**K3**
2.	Determine techniques to gather, differentiate, and document functional and non-functional software requirements.	**K3**
3.	Determine a software project through effort estimation, task scheduling, resource allocation, and quality, risk, and monitoring planning.	**K3**
4.	Apply software design principles and design patterns to address specific software design challenges.	**K3**
5.	Use testing techniques to ensure software quality and reliability during the software implementation process.	**K3**

SYLLABUS

UNIT-I (10Hrs)	The software problem, Process and Project, Component Software Process, Software Development Process Models, Project Management Process
UNIT-II (10 Hrs)	**Software Requirements Analysis and Specification:** Value of a Good SRS, Requirement Process, Requirement Specification, Functional specification with use cases, other approaches for analysis, validation
UNIT-III (10 Hrs)	**Planning a software project:** Effort estimation, project schedule and staffing, quality planning, risk management planning, project monitoring plan, detailed scheduling
UNIT-IV (10 Hrs)	**Software architecture**: Role of software architecture, architecture views, component and connector view, architecture styles for C&C view, documenting architecture design **Design**: design concepts, function-oriented design, object-oriented design, detailed design, verification, metrics

UNIT-V (10 Hrs)	**Coding and unit testing**: Programming principles and guidelines, incrementally developing code, managing evolving code, unit testing, code inspection, metrics. **Testing:** Testing concepts, testing process, Black-box testing, White-box testing, metrics
Textbooks:	
1.	Software Engineering A practitioner's Approach, Roger S. Pressman, 9th Edition, Mc-Graw Hill International Edition.
2.	A Concise Introduction to Software Engineering, Pankaj Jalote, Springer, 2008
Reference Books:	
1.	Software Engineering, Ian Sommerville,10th Edition, Pearson.
2.	Fundamentals of Software Engineering, Rajib Mall, 5th Edition,PHI
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/cec20_cs07/preview

Course Code	Category	L	T	P	C	IM	EM	Exam
B23CEOE07	OE	3	---	---	3	30	70	3 hrs.

ALTERNATIVEENERGY SOURCES

(Offered by CE)

(Offered to AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)

Course Objectives:

1	Explain the concepts of non-renewable and renewable energy systems
2	Outline utilization of renewable energy sources for both domestic and industrial applications
3	Analyze the environmental and cost economics of renewable energy sources in comparison with fossil fuels.

Course Outcomes: After completion of the course, the student will be able to

S. No.	Outcome	Knowledge Level
1	**Summarize** the need of renewable sources in Global scenario	K2
2	**Explain** the solar thermal conversion processes	K2
3	**Explain** the wind energy conversion techniques	K2
4	**Explain** the biomass energy conversion methodologies	K2
5	**Explain** the principle of ocean thermal energy conversion system	K2

SYLLABUS

UNIT-I (10 hrs.)	**Global and National Energy Scenario:** Overview of conventional & renewable energy sources - need & development of renewable energy sources - Global & Indian Energy Scenario - Energy for sustainable development -Globalclimatechange&CO2reduction-concept of Hybrid systems.
UNIT-II (10 hrs.)	**Solar Energy:** Solar energy system-Solar Radiation, Availability, Measurement and Estimation -Solar Thermal Conversion Devices and Storage Applications – Solar Photo voltaic Conversion, applications of solar energy systems.
UNIT-III (10 hrs.)	**Wind Energy:** Wind Energy Conversion - Wind Energy Basics & Characteristics - Wind Turbines (Types & Working) - Site Selection & Wind Resource Assessment - Power Generation & Control - Applications & Safety Aspects - Wind Energy in India.
UNIT-IV (10 hrs.)	**Biogas:** Calorific value and composition of biogas – Bio energy systems – Biomass conversion processes – Thermo chemical conversion processes – biomass gasification – pyrolysis–liquefaction–anaerobicdigestion–Urbanwastetoenergyconversion–bio Diesel production –Biomass energy programme in India.

UNIT-V (10 hrs.)	**Ocean & Geothermal Energy**– Principle of Ocean Thermal Energy Conversion (OTEC) – tidal energy conversion – Scheme of development of tidal energy - Hydro power plants-types of turbines-Geothermal Energy–Geothermal power plants.
Text Books:	
1	Non-Conventional Energy Sources by G.D.Rai
2	Twidell, J.W. and Weir, A., Renewable Energy Sources, EFN Spon Ltd., 1986.
Reference Books:	
1	Kishore V V N, Renewable Energy Engineering and Technology, TeriPress,NewDelhi,2012
2	Godfrey Boyle, Renewable Energy, Power for a Sustainable Future, Oxford University Press, U.K.,1996.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CEOE07	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

WATER SUPPLY SYSTEMS

(Offered by CE)

(offered to AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)

Course Objectives:

1.	To understand the importance of water and analyze various types of water demands in domestic, public, and industrial sectors.
2.	To study different sources of water and methods of collection, treatment, and sustainable utilization.
3.	To learn the principles and components of water supply systems, including storage and distribution methods.
4.	To analyze water quality requirements, wastewater characteristics, and their impact on public health and the environment.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	**Explain** the importance of water and analyze various types of water demands including domestic, public, and industrial uses.	K2
2.	**Describe** different sources of water and methods of water collection, desalination, and groundwater recharge.	K2
3.	**Explain** the concepts of potable and non-potable water, dual water supply systems, and water-related diseases.	K2
4.	**Apply** principles of water distribution systems including storage, conveyance, and supply methods.	K3
5.	**Describe** industrial water requirements, wastewater characteristics, and standards for disposal of industrial effluents.	K2

SYLLABUS

UNIT-I (8Hrs)	**WATER AND LIFE:** Necessity of water – Domestic demand – Public demand – Irrigation – Transportation – Sanitation – Dilution of waste waters – Dust palliative – Recreation – Fire protection.
UNIT-II (8 Hrs)	**SOURCES OF WATER:** Surface sources – Ground sources – Water from atmosphere – Desalination – Recycling of waste water – Recharging of aquifers.
UNIT-III (8 Hrs)	**DUAL SUPPLY OF WATER:** Potable and non-potable water – Protected water – Grey water – Black water – Water bornediseases – water related diseases – Sewage Irrigation.

UNIT-IV (10 Hrs)	**DISTRIBUTION OF WATER:** Based on topography – Gravity distribution – Direct pumping – Combined pumping and gravity flow. Service Reservoirs – Continuous supply – Intermittent supply – Networks of distribution– Emergency water supply as in case of fire accidents – Valves, hydrants and meters.
UNIT-V (8 Hrs)	**INDUSTRIAL WATER:** Location of Industry with reference to surface sources of water – Quality of water required for industrial operations – characteristics of waste water produced – Standards for letting industrial effluents into sources of water.

Textbooks:	
1.	K.N. Duggal, “Elements of Environmental Engineering”, 7thEdition, S. Chand Publishers, 2010.
2.	Howard S. Peavy, Donand P. Rowe, George Technobanoglous, “Environmental Engineering”, 1stEdition Mc Graw –Hill Publications, Civil Engineering Series, 1985.
Reference Books:	
1.	Hammer and Hammer “Water and wastewater Technology”, 4th Edition, Prentice hall of India, 2003.
2.	B.C.Punmia, “Water Supply Engineering”, Vol. 1, “Waste water Engineering Vol. II”, 2nd Edition, Ashok Jain &Arun Jain, Laxmi Publications Pvt.Ltd, New Delhi, 2008.
3.	Fair, Geyer and Okun, “Water and Waste Water Engineering”, 3rdEdition, Wiley, 2010.
4.	Metcalf and Eddy, “Waste Water Engineering”, 3rd Edition, Tata Mc Graw Hill, 2008.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc22_ce07/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CBOE05	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

PRINCIPLES OF OPERATING SYSTEMS

(Offered by CSBS)

(Offered to CE, ECE, EEE& ME)

Course Objectives: This course is designed to:

1.	Understand the basic concepts and principles of operating systems, including process management, memory management, file systems, and Protection.
2.	Make use of process scheduling algorithms and synchronization techniques to achieve better Performance of a computer system.
3.	Illustrate different conditions for the deadlock and their possible solutions.

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Describe** Operating System features and system structures.	**K2**
2.	**Apply** CPU Scheduling Algorithms for Multi Process and Multi-Threaded Operating systems.	**K3**
3.	**Solve**ProcessSynchronizationproblemstoavoidtheoccurrenceofDeadlock situations	**K3**
4.	**Apply** various Memory Management Schemes for Primary and Secondary memory	**K3**
5.	**Describe** file Operations and protection methods.	**K2**

SYLLABUS

UNIT-I (10Hrs)	**Operating Systems Overview**: Introduction, Operating system functions/Operating systems operations, Types of Operating Systems, Free and Open-Source Operating Systems **System Structures**: Operating System Services, User and Operating-System Interface, system calls, Types of System Calls, system programs, Operating system Design and Implementation, Operating System Structure, Building and Booting on Operating System,Operating system debugging.

UNIT-II **(8 Hrs)**	**Processes**: Process Concept, Process scheduling, Operations on processes, Inter process communication. **Threads and Concurrency**: Multi threading models, Thread libraries, Threading issues. **CPU Scheduling**: Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling.
UNIT-III **(10 Hrs)**	**Synchronization Tools**: The Critical Section Problem, Peterson‟s Solution, Mutex Locks, Semaphores, Monitors, Classic problems of Synchronization. **Deadlocks:** system Model, Deadlock characterization, Methods for handling Deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlock‟s
UNIT-IV **(10 Hrs)**	**Memory-Management Strategies**: Introduction, Contiguous memory allocation, Paging, Structure of the Page Table, Swapping. **Virtual Memory Management:** Introduction, Demand paging, copy on-write, Page replacement, Frame Allocation, Thrashing, **Storage Management:** Overview of Mass Storage Structure, HDD Scheduling
UNIT-V **(8 Hrs)**	**File System**: FileSystemInterface: Fileconcept, Access methods, DirectoryStructure. **File system Implementation**: File-systemstructure, File-system Operations, Directory implementation, Allocation method, Free space management. **System Protection**: Goals of protection, Principles and domain of protection, Access matrix, Access control, Revocation of access rights.
Textbooks:	
1.	OperatingSystemConcepts,SilberschatzA,GalvinPB,GagneG,10thEditionWiley,2018
2.	Modern Operating Systems, Tanenbaum AS, 4thEdition,Pearson, 2016.
Reference Books:	
1.	Operating Systems-Internals and Design Principles, Stallings W,9thedition, Pearson,2018.
2.	Operating Systems: A Concept Based Approach, D.M Dhamdhere,3rd Edition, McGrawHill, 2013.
3.	Nutt G, Operating Systems,3rd edition, Pearson Education,2004.
4.	OperatingSystems-InternalsandDesignPrinciples,StallingsW,9thedition, Pearson,2018.
5.	Operating Systems: A Concept Based Approach, D.M Dhamdhere,3rd Edition, McGrawHill, 2013.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CSOE05	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

PRINCIPLES OF SOFTWARE ENGINEERING

(Offered by CSE)

(Offered to CE, ECE, EEE & ME)

Course Objectives: The main objectives of the course is to

1.	Give exposure to phases of software development and common process models.
2.	Give exposure to a variety of software engineering practices such as requirements analysis and specification
3.	Give exposure to software planning strategies
4.	Give exposure to software design techniques
5.	Give exposure to various software testing methods and strategies

Course Outcomes: At the end of the course Students will be able to

S.No	Outcome	Knowledge Level
1.	**Demonstrate** the key concepts of software engineering and software development process models in real-world scenarios.	K3
2.	**Determine** techniques to gather, differentiate, and document functional and non-functional software requirements.	K3
3.	**Determine** a software project through effort estimation, task scheduling, resource allocation, and quality, risk, and monitoring planning.	K3
4.	**Apply** software design principles and design patterns to address specific software design challenges.	K3
5.	**Use** testing techniques to ensure software quality and reliability during the software implementation process.	K3

SYLLABUS

UNIT-I (10Hrs)	The software problem, Process and Project, Component Software Process, Software Development Process Models, Project Management Process
UNIT-II (10 Hrs)	**Software Requirements Analysis and Specification:** Value of a Good SRS, Requirement Process, Requirement Process, Requirement Specification, Functional specification with use cases, other approaches for analysis, validation
UNIT-III (10 Hrs)	**Planning a software project:** Effort estimation, project schedule and staffing, quality planning, risk management planning, project monitoring plan, detailed scheduling
UNIT-IV	**Software architecture**: Role of software architecture, architecture views, component and

(10 Hrs)	connector view, architecture styles for C&C view, documenting architecture design **Design**: design concepts, function-oriented design, object-oriented design, detailed design, verification, metrics
UNIT-V (10 Hrs)	**Coding and unit testing**: Programming principles and guidelines, incrementally developing code, managing evolving code, unit testing, code inspection, metrics. **Testing:** Testing concepts, testing process, Black-box testing, White-box testing, metrics
Textbooks:	
1.	Software Engineering A Practitioner's Approach, Roger S. Pressman, 9th Edition, Mc-Graw Hill International Edition.
2.	A Concise Introduction to Software Engineering, Pankaj Jalote, Springer, 2008
Reference Books:	
1.	Software Engineering, Ian Sommerville,10th Edition, Pearson.
2.	Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/cec20_cs07/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CSOE06	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

COMPUTER NETWORKS

(Offered by CSE)

(Offered to CE,ECE,EEE,&ME)

Course Objectives: The main objectives of the course is to

1.	To understanding the principles of computer networks.
2.	To familiarize with Reference model OSI and TCP/IP
3.	To explore Datalink, Transport and Network layer protocols
4.	To study application layer applications

Course Outcomes: At the end of the course Students will be able to

S. No	Outcome	Knowledge Level
1.	**Summarize** the fundamentals of computer networks and reference models, and analyze their design approaches.	K2
2.	**Apply** data link layer protocols, error detection mechanisms, and multiple access techniques in LAN setups.	K3
3.	**Explain** network layer functions including routing, congestion control, and IP addressing schemes.	K2
4.	**Develop** transport layer services and compare the working of UDP, TCP, and SCTP protocols.	K3
5.	**Demonstrate** application layer protocols for web communication, domain resolution, and file transfers.	K3

SYLLABUS

UNIT-I (10Hrs)	**Introduction:** Types of Computer Networks, Network Topologies Reference Models- The OSI Reference Model, The TCP/IP Reference Model, A Comparison of the OSI and TCP/IP Reference Models. **Physical Layer**: Introduction to physical layer, Guided Media- Twisted-pair cable, Coaxial cable and Fiber optic cable and unguided media
UNIT-II (10 Hrs)	**The Data Link Layer:** Data Link Layer Design Issues, Services Provided to the Network Layer, Error detecting and Error Correcting codes, Elementary Data Link Protocols, Sliding Window Protocols, HDLC. Multiple Access Protocols in Wired Lans, Ethernet, Fast Ethernet, Gigabit Ethernet
UNIT-III (10 Hrs)	**The Network Layer**: Network Layer Design Issues, Routing Algorithms, Congestion, Congestion control algorithms. The Network Layer in the Internet, The IP Version 4

	Protocol, IP Addresses- Classful, CIDR, NAT, IP Version 6 Protocol, Transition from IPV4 to IPV6
UNIT-IV (10 Hrs)	**The Transport Layer:** The Transport Layer Services, Connection Establishment and Termination, Congestion Control, Sliding Window Protocol, Transport Layer Protocols: UDP, TCP and SCTP
UNIT-V (10 Hrs)	**The Application Layer:** Services And Protocols, The World Wide Web, HTTP, Domain Name Space, Remote Loging, Electronic Mail and File Transfer

Textbooks:	
1.	"Computer Networks", Andrew S Tanenbaum, David J Wetherall, 5^{th} Edition, Pearson
2.	"Data Communications and Networking", Behrouz A Forouzan, 4^{th} Edition, Tata McGraw Hill Education
Reference Books:	
1.	"Data and Computer Communication", William Stallings, Pearson
2.	"TCP/IP Protocol Suite", Behrouz Forouzan, McGraw Hill.
e-Resources	
1.	https://nptel.ac.in/courses/106105183

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CIOE07	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

INTODUCTION TO CYBER SECURITY

(Offered by CIC)

(Offered to CE,ECE,EEE&ME)

Course Objectives: The Objective of the course is to

1.	identify security risks and take preventive steps
2.	understand the forensics fundamentals
3.	understand the evidence capturing process
4.	understand the preservation of digital evidence

Course Out Comes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Apply knowledge of cybercrime concepts to identify different types of cybercrimes and basic attacks on computers, networks, and mobile devices.	K3
2.	Analyze various cyber attack tools and techniques such as phishing, malware, spoofing, session hijacking, SQL injection, DoS/DDoS attacks, wireless network attacks	K4
3.	Analyze cybercrime investigation techniques including digital evidence collection, e-mail and IP tracking.	K4
4.	Apply computer forensics tools and techniques to investigate digital evidence from computer systems, networks, e-mails, and mobile devices	K3
5.	Apply the concepts of cyber law and the Indian IT Act to understand cybercrime issues, legal provisions, and punishments in the Indian cyber security scenario.	K3

SYLLABUS

UNIT-I (10 Hrs)	Introduction to Cybercrime: Introduction, Cybercrime: Definition and Origins of the Word, Cybercrime and Information Security, Cybercriminals, Classifications of Cybercrime, Cyberstalking, Cybercafe and Cybercrimes, Botnets. Attack Vector, Proliferation of Mobile and Wireless Devices, Security Challenges Posed by Mobile Devices, Attacks on Mobile/Cell Phones, Network and Computer Attacks.
UNIT-II (10 Hrs)	Tools and Methods : Proxy Servers and Anonymizers, Phishing, Password Cracking, Keyloggers and Spywares, Virus and Worms, Trojan Horses and Backdoors, Steganography, Sniffers, Spoofing, Session Hijacking Buffer over flow, DoS and DDoS Attacks, SQL Injection, Buffer Overflow, Attacks on Wireless Networks, Identity Theft (ID Theft), Foot Printing and Social Engineering, Port Scanning, Enumeration.

UNIT-III (10 Hrs)	Cyber Crime Investigation: Introduction, Investigation Tools, eDiscovery, Digital Evidence Collection, Evidence Preservation, E-Mail Investigation, E-Mail Tracking, IP Tracking, E-Mail Recovery, Hands on Case Studies. Encryption and Decryption Methods, Search and Seizure of Computers, Recovering Deleted Evidences, Password Cracking.
UNIT-IV (10 Hrs)	Computer Forensics and Investigations: Understanding Computer Forensics, Preparing for Computer Investigations. Current Computer Forensics Tools: Evaluating Computer Forensics Tools, Computer Forensics Software Tools, Computer Forensics Hardware Tools, Validating and Testing Forensics Software, Face, Iris and Fingerprint Recognition, Audio Video Analysis, Windows System Forensics, Linux System Forensics, Graphics and Network Forensics, E-mail Investigations, Cell Phone and Mobile Device Forensics.
UNIT-V (10 Hrs)	Cyber Crime Legal Perspectives: Introduction, Cybercrime and the Legal Landscape around the World, The Indian IT Act, Challenges to Indian Law and Cybercrime Scenario in India, Consequences of Not Addressing the Weakness in Information Technology Act, Digital Signatures and the Indian IT Act, Amendments to the Indian IT Act, Cybercrime and Punishment, Cyberlaw, Technology and Students: Indian Scenario

TEXTBOOK:

1.	Sunit Belapure Nina Godbole "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", WILEY, 2011.
2.	Nelson Phillips and Enfinger Steuart, "Computer Forensics and Investigations", Cengage Learnir New Delhi, 2009.

REFERENCE BOOKS:

1.	Michael T. Simpson, Kent Backman and James E. Corley, "Hands on Ethical Hacking and Network Defence", Cengage, 2019.
2.	Computer Forensics, Computer Crime Investigation by John R. Vacca, Firewall Media, New Delhi.
3.	Alfred Basta, Nadine Basta,Mary Brown and Ravinder Kumar "Cyber Security and Cyber Laws" , Cengage,2018.
4.	Computer Forensics, Computer Crime Investigation by John R. Vacca, Firewall Media, New Delhi.

E-Resources:

1.	https://www.netacad.com/courses/introduction-to-cybersecurity?courseLang=en-US
2.	CERT-In Guidelines- http://www.cert-in.org.in/
3.	https://www.coursera.org/learn/introduction-cybersecurity-cyber-attacks [Online Course]
4.	https://computersecurity.stanford.edu/free-online-videos [Free Online Videos]
5.	Nickolai Zeldovich. 6.858 Computer Systems Security. Fall 2014. Massachusetts Institute of Technology: MIT Open Course Ware, https://ocw.mit.edu License: Creative Commons BY-NC-SA.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CIOE08	OE	3	--	--	3	30	70	3 Hrs.

INTRODUCTION TO BLOCKCHAIN TECHNOLOGIES

(Offered by CIC)

(Offered to CE,ECE,EEE&ME)

Prerequisites: Cryptography & Network Security

Course Objectives:

1.	To learn the fundamentals of Block Chain and various types of block chain and consensus mechanism.
2.	To understand public block chain system, Private block chain system and consortium block chain.
3.	Able to know the security issues of blockchain technology.

Course Outcomes

S.No	Outcomes	Knowledge Level
1.	Explain the fundamental concepts of the Block chain technology.	K2
2.	Demonstrate public block chain system concepts and the smart contracts.	K2
3.	Demonstrate private and consortium block chain systems and Initial Coin Offering.	K2
4.	Apply blockchain concepts and security to the different applications.	K3
5.	Develop block chain programs and chain codes using python and Hyperledger fabric	K3

SYLLABUS

UNIT-I (10Hrs)	**Fundamentals of Blockchain:** Introduction, Origin of Blockchain, Blockchain Solution, Components of Blockchain, Block in a Blockchain, The Technology and the Future. **Blockchain Types and Consensus Mechanism:** Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. **Cryptocurrency:** Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.
UNIT-II (10 Hrs)	**Public Blockchain System:** Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. **Smart Contracts:** Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.
UNIT-III (10 Hrs)	**Private Blockchain System:** Introduction, Key Characteristics of Private Blockchain, Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-

	commerce Site Example, Smart Contract in Private Environment. **Consortium Blockchain:** Introduction, Key Characteristics of Consortium Blockchain, Need of Consortium Blockchain, Hyperledger Platform. **Initial Coin Offering:** Introduction, Blockchain Fundraising Methods, Launching an ICO, Investing in an ICO, Pros and Cons of Initial Coin Offering.
UNIT-IV (10 Hrs)	**Security in Blockchain**: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric. **Applications of Blockchain:** Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.
UNIT-V (10 Hrs)	**Blockchain Platform using Python:** Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. **Blockchain platform using Hyperledger Fabric:** Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, **Blockchain Case Studies:** Retail, Banking and Financial Services.

Textbooks:	
1.	"Block chain Technology", Chandramouli Subramanian, Asha A.George, Abhilasj K A and Meena Karthikeyan, Universities Press, First Edition, 2020.
Reference Books:	
1.	Blockchain Blue print for Economy, Melanie Swan, SPD Oreilly.
2.	Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gauar, Pearson Addition Wesley.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ECOE06	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

PRINCIPLES OF CELLULAR & MOBILE COMMUNICATIONS

(Offered by ECE)

(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT &ME)

Course Objectives: Students are expected to

1.	Understand the fundamentals of cellular communication systems, including limitations of conventional systems, types of interferences, and handoff mechanisms.
2.	Apply concepts of frequency management, channel assignment, and multiple access techniques in mobile communication systems.
3.	Develop skills to explore, analyze, and present emerging trends and technologies in mobile and cellular communications.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** cellular system operation and design.	**K3**
2.	**Analyze** co-channel and non-co-channel interference and diversity techniques.	**K3**
3.	**Understand** channel assignment and frequency management techniques.	**K2**
4.	**Explain** handoff types and their impact on system performance.	**K2**
5.	**Understand** Multiple access schemes in mobile and to develop the ability to search, select, organize and present information on new technologies in mobile and cellular communications	**K3**

SYLLABUS

UNIT-I (12Hrs)	**CELLULAR SYSTEMS:** Limitations of Conventional System, Basic Cellular Mobile System, First, second, third and fourth and Generation cellular wireless systems. Operation of Cellular System. Uniqueness of Mobile Radio Environment –Fading, coherence bandwidth, Doppler Spread. Fundamentals of cellular Radio System Design: concept of frequency reuse channels, Co-channel Interference, Co-channel Interference Reduction Factor, desired C/I from a normal case in a Omni directional Antenna system. Trunking and grade of service
UNIT- II (10 Hrs)	**CO-CHANNEL & NON-CO-CHANNEL INTERFERENCE:** Measurement of Real Time Co-Channel Interference, design of Antenna system, Antenna parameters and their effects, diversity techniques: Space Diversity, Polarization diversity, frequency diversity and time diversity. Non-co channel interference-adjacent channel interference.
UNIT- III (10 Hrs)	**FREQUENCY MANAGEMENT AND CHANNEL ASSIGNMENT:** Numbering and grouping, setup access and paging channels, channel assignments to cell sites and mobile

	units, channel sharing and borrowing, sectorization, overlaid cells, non-fixed channel assignment.
UNIT-IV (10 Hrs)	**HANDOFFS:** Handoff Initiation, types of handoff, delaying handoff, advantages of Handoff, power difference handoff, forced handoff, mobile assisted and soft and Hard handoffs. Intersystem and Intrasystem handoffs, Concept of Call drop rates.
UNIT- V (08 Hrs)	**DIGITAL CELLULAR NETWORKS AND MULTIPLE ACCESS SCHEME:** GSM architecture, GSM channels, TDMA, FDMA, and CDMA. Introduction to MIMO systems Principles of CDMA cellular systems, Introduction of 5G.

Textbooks:	
1.	Mobile Cellular Telecommunications – W.C.Y. Lee, Tata McGraw Hill, 2rd Edn., 2006. Wireless Communications - Theodore. S. Rapport, Pearson education, 2nd Edn., 2002.
2.	T.S.Rappaport,“WirelessCommunications–PrinciplesandPractice”(2ndedition) Pearson, 2010, ISBN 9788131731864.
Reference Books:	
1.	Principles of Mobile Communications–GordonL. Stuber, Springer International 2nd Edition, 2001
2.	Modern Wireless Communication–Simon Haykin Michael Moher, Persons Eduction, 2005.3. Wireless Communication theory and Techniques, Asrar U.H. Sheikh Springer, 2004
3.	A.Goldsmith,“WirelessCommunications,”CambridgeUnivPress,2005
4.	D.TseandP.Viswanath,“FundamentalsofWirelessCommunications,”CambridgeUniv Press, 2005
e-Resources	
1.	https://www.youtube.com/watch?v=f2wlHL1Sok8&list=PLuv3GM6gsE3ypUYh43pPuZsXxJVG1e7F
2.	https://www.youtube.com/watch?v=whYljse4Abc
3.	https://archive.nptel.ac.in/courses/128/108/128108016/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ECOE07	OE	3	--	--	3	30	70	3 Hrs.

IMAGE PROCESSING

(Offered by ECE)

(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT &ME)

Course Objectives: Students are expected

1.	To introduce fundamentals of digital image processing and study image transforms
2.	To demonstrate digital image processing techniques in spatial and frequency domains
3.	To study advanced image analysis methods: image segmentation, morphological image processing

Course Outcomes: At the end of the course Students will be able to

S.No	Outcome	Knowledge Level
1.	**Understand** the basic principles of digital image processing and perform image transforms	K2
2.	**Perform** basic image processing methods such as Image filtering operations, Image enhancement.	K3
3.	**Analyze** and compare various image compression techniques and their applications	K4
4.	**Design** and implement various algorithms for image analysis	K4

SYLLABUS

UNIT-I (10Hrs)	**Fundamentals of Image Processing:** Digital Image Fundamentals, Basic steps of Image Processing System, Sampling and Quantization of an image, relationship between pixels, Imaging Geometry.
UNIT- II (10 Hrs)	**Image Enhancement:** Spatial domain methods: Histogram processing, Fundamentals of Spatial filtering, Smoothing spatial filters, Sharpening spatial filters. Frequency domain methods: Basics of filtering in frequency domain, image smoothing, image sharpening, Selective filtering.
UNIT- III (10 Hrs)	**Image Segmentation:** Segmentation concepts, Point, Line and Edge Detection, Edge Linking using Hough Transform, Thresholding, Region Based segmentation. Wavelet based Image Processing: Introduction to wavelet Transform.
UNIT-IV (10 Hrs)	**Image Compression:** Image compression fundamentals - Coding Redundancy, Spatial and Temporal redundancy, Compression models: Lossy and Lossless, Huffman coding, Arithmetic coding.

UNIT- V (10 Hrs)	**Image Restoration:** Image Restoration Degradation model, Algebraic approach to restoration, Inverse Filtering. Morphological Image Processing: Dilation and Erosion, Opening and closing, the hit or miss Transformation, some basic morphological algorithms.
Textbooks:	
1.	Digital Image Processing, Rafael C. Gonzalez and Richard E.Woods, 4thEdition, Pearson, 2018
2.	Digital Image Processing, S.Jayaraman, S. Esakkirajan, T. Veerakumar, 5th Edition, TMH, 2015
Reference Books:	
1.	Digital Image Processing, William K.Pratt, 3rdEdition, John Willey, 2007
2.	Fundamentals of Digital Image Processing, A.K.Jain, 3rd Edition, PHI, 1989
e-Resources	
1.	**https://onlinecourses.nptel.ac.in/noc19_ee55/preview**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ECOE08	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

FUNDAMENTALS OF SATELLITE COMMUNICATIONS

(Offered by ECE)

(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT &ME)

Course Objectives:

1.	An ability to apply the fundamentals of keplers planetary motion in satellite communication and GPS.
2.	An ability to analyze the space segment of satellite communication systems and multiple access Techniques
3.	An ability to design link parameters of satellite uplink and downlink budget.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** fundamentals of Keplers planetary motion in satellite communication and GPS.	**K3**
2.	**Analyze** and build the space segment, depending upon the requirement.	**K3**
3.	**Design** link margin for various applications.	**K3**
4.	**Choose** the correct multiple access technique for better communication	**K2**
5.	**Describe** GNSS fundamentals and identify major constellations including GPS, GLONASS and Galileo	**K3**

SYLLABUS

UNIT-I (10Hrs)	**INTRODUCTION:** Basic Concepts of Satellite Communications, Frequency allocations for Satellite Services, Applications, and Future Trends of Satellite Communications. Orbital Mechanics, launches and launch vehicles, Orbital effects in communication systems performance
UNIT- II (10 Hrs)	**SATELLITE SUBSYSTEMS:** Attitude and orbit control system, telemetry, tracking, Command and monitoring system, power systems, communication subsystems, Satellite antennas.
UNIT- III (10 Hrs)	**SATELLITE LINK DESIGN:** Basic transmission theory, link equation, C/N ratio, system noise temperature and G/T ratio, Design of satellite link.
UNIT-IV (10 Hrs)	**MULTIPLE ACCESS:** Frequency division multiple access (FDMA): Intermodulation, Calculation of C/N. Time division Multiple Access (TDMA); Frame structure, Examples. Code Division Multiple access (CDMA): Spread spectrum transmission and reception.

UNIT- V (10 Hrs)	**GLOBAL NAVIGATION SATELLITE SYSTEM (GNSS):** Introduction, various GNSS: GPS, GLONASS, GALILEO, BeiDou, IRNSS. GPS-location principle, GPS navigation message, GPS receiver operation, differential GPS.
Textbooks:	
1.	Satellite Communications – Timothy Pratt, Charles Bostian and Jeremy Allnutt, WSE, Wiley Publications,3RD Edition, 2020.
2.	Satellite Communications – Dennis Roddy, McGraw Hill, 2nd Edition
Reference Books:	
1.	Satellite Communication Systems Engineering, by Wilbur L. Pritchard, Henri G. Suyderhoud, Robert A.Nelson (Second Edition), Pearson
2.	Global Navigation Satellite Systems with Essentials of Satellite Communication—G.S.RAO
e-Resources	
1.	**https://www.isro.gov.in/applications/satellite-communication**
2.	**https://www.isro.gov.in/spacecraft**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EEOE07	OE	3	--	--	3	30	70	3 Hrs.

INTRODUCTION TO DRONE TECHNOLOGY

(Offered by EEE)

(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT & ME)

Course Objectives: Students will learn about

1.	The basic concepts of Drone, its classification and Safety essentials.
2.	The knowledge on the hardware components of Quadcopter.
3.	The knowledge of aero dynamic lift, design and assembly of drone.
4.	The knowledge on drone sensors and Indicators.
5.	The knowledge to develop Pluto Drone and its App Communication.

Course Outcomes: At the end of the course, the students will be able to

S.No	Outcome	Knowledge Level
1.	Explore the history, types and safety essentials of Drone.	K3
2.	Identify various components of Quadcopter.	K3
3.	Illustrate the aero dynamic lift of Quadcopter.	K3
4.	Illustrate the Drone sensor technology and Indicators.	K3
5.	Assemble of Pluto Drone and flying with Pluto App Communication.	K4

SYLLABUS

UNIT-I (10Hrs)	**Introduction to Drones** Introduction to Drones and history of evolution of Drones, Types of Drones based on structure-Single Rotor Helicopter, Multirotor Drones, Fixed Wing Drones, Fixed Wing Hybrid VTOL Drones, Lighter Than Air Systems (LTA) Drones, Flappy Wing Drones, Classification of Drones in India based on maximum all-up weight including payload. Applications of Drones in various fields. **Drone Safety Essentials** Pre-Flight Checklist, In-Flight Check list, Post Flight Check List, Safety and Mandates in Flying Drones, Do’s and Don’ts while operating Drones.
UNIT-II (10 Hrs.)	**Components of a Quadcopter** Flight Controller, Frame, Motors-Brushed Direct Current (DC) motors and Brushless Direct Current (BLDC) motors, Electronic Speed Controller, Propellers, Propellers Guards, Battery-Lithium Polymer (LiPo) Battery, Radio Transmitter and Receiver, General Block Diagram of Multirotor Drone.
UNIT-III	**Aerodynamics of a Quadcopter**

(10 Hrs.)	Principal Axes and Rotation of Drone, Fundamental Movement and controls, Motor Configuration of a Quadcopter, Propeller Configuration, Forces acting on a Quadcopter, Propulsion System of Multirotor Drones, Torque or Moments acting on Quadcopter, Quad copter's Flight Dynamics.
UNIT-IV (10 Hrs.)	**Drone Sensors and Indicators** Need of Sensors on Drones, Fundamental In-Built Sensors-Accelerometer, Barometer, Gyro Sensor, Magnetometer, Add-on Sensors-GPS, Proximity Sensors, Thermal Cameras, Multispectral Sensors, Indicators on Drone.
UNIT-V (10 Hrs.)	**Building Pluto Drone** Components in Pluto Drone Ecosystem, Steps to Build Pluto Drone, Flying with Pluto Controller App-Download Pluto Controller App, Connect to Drone, App User Interface, Sensors Calibration-Accelerometer Calibration and Magnetometer Calibration, Motor Test, Update Firmware, Understanding Controls, Arm and Fly, Control settings, Trim Settings, Developer Mode.
Textbooks:	
1.	I.V.S. Yeswanth, Dr A.V.S. Sridhar Kumar "Fundamentals of Drone Technology" , Authors click publishing, First Edition 2024.
2.	Dharna Nar, Radhika Kotecha "Drone Technology for Beginners", Drones School India, First Edition 2024.
Reference Books:	
1.	Garg PK, "Unmanned aerial vehicles: An introduction", Mercury Learning and Information; 2021
2	Paul G Fahlstrom, Thomas J Gleason, "Introduction to UAV Systems", UAV Systems, Inc,1998.
e-Resources:	
1.	ATAL-Drone Module: **https://aim.gov.in**
2.	https://onlinecourses.nptel.ac.in/noc25_ae30/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EEOE08	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

ENERGY MANAGEMENT AND AUDIT

(Offered by EEE)

(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT & ME)

Course Objectives: Students will learn about

1.	Energy scenario in India
2.	Principles of Energy Management, Qualities and functions of energy manager.
3.	Waste Heat Recovery, Steam Generation, Distribution, Utilization: Methodology
4.	Power factor control, Tariff, Energy Efficient Lighting, Life Cycle Cost Analysis
5.	Objectives and types of Energy Audit, Energy Conservation Schemes

Course Outcomes: At the end of the course, the students will be able to

S.No	Outcome	Knowledge Level
1.	Explain the fundamental aspects of energy scenario in India	K3
2.	Explain Principles of Energy management and various duties of energy manager	K3
3.	Illustrate the Waste heat recovery	K3
4.	Design energy efficient lighting systems and explain the power factor control	K3
5.	Explain about the methods and concept of energy audit	K3

SYLLABUS

UNIT-I (10Hrs)	**Energy Scenario**: Introduction, Classification of Energy – Primary and Secondary Energy, Commercial Energy and Non-commercial Energy and Renewable & Non-renewable energy; Various forms of Energy – potential (stored energy) and kinetic (working) energy; Energy reserves and production; Energy consumption, Energy units & conversions.
UNIT-II (10 Hrs)	**Energy Management:** Introduction, Scope of energy management, Necessary steps of energy management programme, Principles of Energy Management, Qualities and functions of energy manager, Energy management Act.
UNIT-III (10 Hrs)	**Thermal Energy Management:** Waste Heat Recovery: Waste Heat, Advantages of Recuperators, Heat Pipe Heat Exchanger, Heat wheel, Heat Pumps, Air preheaters and Economizers, Steam Generation, Distribution, Utilization: Methodology.
UNIT-IV	**Electrical Energy Management:**

(10 Hrs)	Introduction, Power factor control, Tariff, Energy efficient motors, Case Study, Energy Efficient Lighting, Life Cycle Cost Analysis, Equivalent Annual Worth (EAW), Break even Analysis.
UNIT-V (10 Hrs)	**Energy Auditing**: Introduction, Objectives of Energy Audit, Control of Energy, Uses of Energy, Energy Conservation Schemes, Energy Index, Cost Index, Pie Chart, Sankey Diagram, Load Profile, Types of Energy Audit, General Energy Audit, Energy Audit Case Studies.

Textbooks:	
1.	Energy Management and Conservation by K.V. Sharma &P.Venkatasesaiah, I.K. International Pvt. Ltd., New Delhi, 2020.
2.	Energy management by W.R. Murphy & G. Mckay Butter worth, Elsevier publications, 2012.
Reference Books:	
1.	Energy management by Paul o' Callaghan, Mc–Graw Hill Book company–1st edition, 1998.
2.	Energy management hand book by W.C.Turner, John wiley and sons.
e-Resources:	
1.	https://onlinecourses.swayam2.ac.in/nou25_es05/preview
2.	https://beeindia.gov.in/en/energy-auditors

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ITOE04	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

PRINCIPLES OF OPERATING SYSTEMS

(Offered by IT)

(Offered to CE, ECE, EEE & ME)

Course Objectives: The aim of the course is to:

1.	Provide a strong foundation in the fundamental concepts and principles of operating systems, process scheduling algorithms and synchronization techniques.
2.	Examine deadlock conditions and explore various techniques for their prevention, avoidance, and resolution
3.	Gain knowledge of memory management strategies such as paging, virtual memory, and allocation techniques
4.	Acquire knowledge of file system structures, implementation, and storage management concepts

Course Outcomes: By the end of the course, student will be able to::

S.No	Outcome	Knowledge Level
1.	**Explain** the structure and functions of operating systems and their role in computer systems	K2
2.	**Apply** process scheduling and synchronization mechanisms to solve concurrency problems	K3
3.	**Analyze** and handle deadlocks using appropriate techniques	K3
4.	**Apply** and assess memory management schemes such as paging and virtual memory.	K3
5.	**Illustrate** file system organization, implementation, and storage management methods	K2

SYLLABUS

UNIT-I	**Operating Systems Overview**: Introduction, Operating system functions, Operating systems operations, Computing environments, Free and Open-Source Operating Systems. **System Structures**: Operating System Services, User and Operating-System Interface, system calls, Types of System Calls, system programs, Operating system Design and Implementation, Operating system structure.
UNIT-II	**Processes:** Process Concept, Process scheduling, Operations on processes, Inter-process communication. **Threads and Concurrency**: Multithreading models, Threading issues.

	CPU Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms.
UNIT-III	**Synchronization Tools:** The critical section problem, Peterson's solution, Mutex Locks, Semaphores, Monitors. **Deadlocks**: System model, Deadlock characterization, Methods for handling deadlocks.

UNIT-IV	**Memory-Management Strategies**: Introduction, Contiguous memory allocation, Paging, Structure of the Page Table, Swapping. **Virtual Memory Management**: Introduction, Demand paging, Copy-on-write, Page replacement. **Storage Management**: Overview of Mass Storage Structure, HDD Scheduling.
UNIT-V	**File System:** File System Interface: File concept, Access methods, Directory Structure. **File system Implementation:** File-system structure, File-system Operations, Directory implementation, Allocation method, Free space management.
Textbooks:	
1.	Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley,2018
2.	Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson, 2016
Reference Books:	
1.	Operating Systems: A Concept Based Approach, D.M Dhamdhere, 3rd Edition, McGraw- Hill, 2013

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23MEOE09	OE	3	--	--	3	30	70	3 Hrs.

ADDITIVE MANUFACTURING

(Offered by ME)

(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE & IT)

Course Objectives:

1.	The course is designed to develop fundamental knowledge of Additive Manufacturing
2.	Study the liquid-based, solid-based, and powder-based Additive Manufacturing techniques
3.	Learn tools used for Additive Manufacturing

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Apply the working principles and process parameters of additive manufacturing processes.	K3
2.	Illustrate various liquid and solid-based additive manufacturing processes.	K3
3.	Illustrate powder-based AM processes, and apply suitable post-processing methods to remove support structures and improve part quality.	K3
4.	Apply reverse engineering techniques to generate CAD models for rapid prototyping.	K3
5.	Apply rapid tooling methods for AM Production.	K3

SYLLABUS

UNIT-I (10Hrs)	**Introduction to AM:** Principle of AM, AM evolution, Generic AM process chain, classification, AM v/s Conventional Machining, Advantages and limitations. **Design for AM:** Preparation of CAD Models – STL File, STL File Format, STL File Problems, Software for Slicing, Newly Proposed Formats, Part Orientation, Support Structure
UNIT-II (10 Hrs)	**Additive Manufacturing Processes:** **Liquid-Based AM:** Stereolithography (SL) – Apparatus, Working Principle, Process Modeling: Process Parameters, photopolymers, photo polymerization, layering technology, advantages, disadvantages, limitations & Applications. case studies. **Solid-Based AM:**Fused Deposition Modelling (FDM), Laminated object Manufacturing (LOM), Ultrasonic AM- Working Principle, process modeling, process parameters, Materials, advantages, Applications, Limitations.
UNIT-III (10 Hrs)	**Powder-Based AM:** Selective Laser Sintering (SLS), Direct Metal Laser Sintering (DMLS) - Working Principle, Processes Modeling, Process Parameters, Advantages,

	Applications, Limitations. Three dimensional printing (3DP): Models and specifications, process, working principle, applications, advantages and disadvantages, **Post Processing Treatment in AM**: Support Material Removal, Improve surface Quality-Property Enhancements using Non-thermal and Thermal Techniques.
UNIT-IV (10 Hrs)	**Reverse Engineering:** Introduction- Digitization techniques – Model Reconstruction – Data Processing for Rapid Prototyping: CAD model preparation, Data Requirements. **Materials available for AM:** Liquid-Based, Solid-Based, Powder-Based.
UNIT-V (10 Hrs)	**Rapid Tooling:** Introduction - Conventional v/s RT – Classification : Direct and Indirect - Direct v/s Indirect tooling - Direct Methods : Laminated Tooling, DMLS - Indirect Methods : RTV EpoxyTools, 3D Keltool - Applications of RT **Application Areas for AM**: Automotive Industry, Aerospace Industry, Medical and Bioengineering applications

Textbooks:	
1.	Additive Manufacturing Technologies, Gibson, Ian, David W. Rosen, Brent Stucker, and Mahyar Khorasani, Springer, 2021
2.	Rapid prototyping: Principles and applications, second edition, Chua C.K., Leong K.F., and Lim C.S., World Scientific Publishers, 2003.
3.	Rapid Tooling: Technologies and Industrial Applications, Peter D.Hilton, Hilton/Jacobs, Paul F.Jacobs, CRC press, 2000.
Reference Books:	
1.	Rapid prototyping, Andreas Gebhardt, Hanser Gardener Publications, 2003.
2.	Rapid Prototyping and Engineering applications: A tool box for prototype development, LiouW.Liou, Frank W.Liou, CRC Press, 2007.
3.	Rapid Prototyping: Theory and practice, Ali K. Kamrani, Emad Abouel Nasr, Springer, 2006.
4.	Paul F.Jacobs – “Stereo lithography and other RP & M Technologies”, SME, NY 2011
5.	Additive Manufacturing Technologies: 3D Printing, Rapid Prototyping, and Direct Digital Manufacturing, Ian Gibson, David W Rosen, Brent Stucker, Springer, 2015, 2nd Edition.
e-Resources	
1.	https://courses.gen3d.com/courses/enrolled/988400
2.	https://all3dp.com/1/design-for-additive-manufacturing-dfam-simply-explained/#where-to
3	https://markforged.com/resources/blog/design-for-additive-manufacturing-dfam

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MEOE10	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

INDUSTRIAL SAFETY

(Offered by ME)

(Offered to AIDS, AIML, CE, CSIT, CSBS, CSG, CSE, CIC, ECE, EEE & IT)

Course Objectives:

1	To understand the concepts of industrial safety and management.
2	To demonstrate the accident preventions and protective equipment.
3	To understand and apply the knowledge of safety acts
4	To have the knowledge about fire prevention and protection systems
5	To understand and apply fire safety principles in buildings

Course Outcomes

S.No	Outcome	Knowledge Level
1	Apply the concepts of industrial safety and management in industrial environments.	K3
2	Demonstrate the working principles of smart machines and smart sensors in industrial applications.	K3
3	Apply IoT technologies in Industry 4.0 to develop systems according to industrial requirements.	K3
4	Apply fire prevention methods and fire protection systems in industries.	K3
5	Use fire safety principles in buildings to ensure safe design and evacuation	K3

SYLLABUS

UNIT-I (10Hrs)	**INTRODUCTION TO THE DEVELOPMENT OF INDUSTRIAL SAFETY AND MANAGEMENT:** History and development of Industrial safety: Implementation of factories act, Safety and productivity, Safety organizations. Safety committees and structure, Role of management and role of Govt. in industrial safety. The ISO 9000 ANSI/ASQCQ- series standards, benefits of ISO 9000 certifications.
UNIT-II (10 Hrs)	**ACCIDENT PREVENTIONS AND PROTECTIVE EQUIPMENT:** Personal protective equipment, Survey the plant for locations, Part of body to be protected, Education and training in safety, Prevention causes and cost of accident, Housekeeping, First aid, Accident reporting, Investigations. Industrial psychology in accident prevention, Safety trials, Safety related to operations.
UNIT-III (10 Hrs)	**SAFETY ACTS:** Features of Factory Act, Introduction of Explosive Act, Boiler Act, ESI Act, Workman's compensation Act, Industrial hygiene, Occupational safety, Diseases

	prevention, Ergonomics, Occupational diseases, stress, fatigue, health, safety and the physical environment, Engineering methods of controlling chemical hazards, safety and the physical environment, Control of industrial noise and protection against it, Code and regulations for worker safety and health, codes for safety of systems.
UNIT-IV (10 Hrs)	**FIRE PREVENTION AND PROTECTION:** Sources of ignition – fire triangle – principles of fire extinguishing – active and passive fire protection systems – various classes of fires – A, B, C, D, E – Fire extinguishing agents – Water, Foam, Dry chemical powder, Carbon-dioxide Halon alternatives Halocarbon compounds – Inert gases, dry powders – types of fire extinguishers – fire stoppers – hydrant pipes – hoses – monitors – fire watchers – layout of stand pipes – fire station – fire alarms and sirens – maintenance of fire trucks – foam generators – escape from fire rescue operations – fire drills – first aid for burns
UNIT-V (10 Hrs)	**BUILDING FIRE SAFETY:** Objectives of fire safe building design. Fire load, fire resistant material and fire testing – structural fire protection – structural integrity – concept of egress design – exit – width calculations – fire certificates – fire safety requirements for high rise buildings.

Textbooks:

1.	Industrial Maintenance Management Srivastava, S.K.- S. Chandand Co.
2.	Occupational Safety Management and Engineering Willie Hammer– PrenticeHall
3	Purandare D.D& Abhay D.Purandare, "Handbook on Industrial Fire Safety" P&A publications, NewDelhi, 2006 .
4	McElroy,Frank.E., "Accident Prevention Manual for Industrial Operations", NSC, Chicago, 1988

Reference Books:

1.	Installation, Servicing and Maintenance Bhattacharya, S.N.-S. Chandand Co.
2.	Jain VK "Fire Safety in Building" New Age International 1996.
3.	Reliability, Maintenance and Safety Engineering by Dr.A. K. Guptha
4.	A Text book of Reliability and Maintenance Engineering by Alakesh Manna

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BSOEO4	**OE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

FUZZY SETS AND FUZZY LOGIC

(offered by EM&H)

(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)

Course Objectives: Students are expected to learn

1	Crisp sets, Fuzzy sets and Fuzzy Union, Fuzzy Intersection of Fuzzy sets
2	Lamda cut for fuzzy relations, Fuzzy tolerance and equivalence relations
3	Fuzzification & Defuzzification to scalars by Centroid method, Centre of sums method, Mean of maxima method.
4	Fuzzy logic and crisp connectives, fuzzy logical connectives.
5	Fuzzy control system, fuzzy clustering and its applications (6th one deleted)

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1	Describe Crispsets, Fuzzy sets and operations of Fuzzy sets.	K3
2	Describe different types of Fuzzy relations.	K3
3	Describe Fuzzification and Defuzzification to scalars by Centroid method, Centre of sums method, Mean of maxima method.	K3
4	Describe Fuzzy logic, Crisp connectives, Fuzzy logical connectives.	K3
5	Apply Fuzzy logic to control systems and know about Fuzzy clustering and its applications.	K3

SYLLABUS

UNIT-I (12Hrs)	**Crisp Sets Vs Fuzzy Sets:** Crisp sets an overview, Concept of fuzziness, the notion of Fuzzy sets, basic concepts of fuzzy sets. **Operations of Fuzzy Sets: Fuzzy** set operations –fuzzy complement, fuzzy union, fuzzy intersection, combinations of operations.
UNIT-II (12Hrs)	**Fuzzy Relations:** Fuzzy Cartesian product, Fuzzy relations, operations on fuzzy relations, properties of fuzzy relations, Lamda cut for fuzzy relations and composition, Fuzzy tolerance, and equivalence relations.
UNIT-III (12Hrs)	**Fuzzification and Defuzzification:** Features of membership function, fuzzification, defuzzification to crispset, Defuzzification to scalars (Centroid method, Centre of sums method, Mean of maxima method).
UNIT-IV	**Fuzzy Logic:** Introduction to fuzzy logic, Crisp connectives vs Fuzzy logical connectives,

(12Hrs)	Approximate reasoning.
UNIT-V (12Hrs)	**Applications of Fuzzy Systems:** Fuzzy Control System, Control System Design Problem, Simple Fuzzy Logic Controller, Fuzzy clustering and its applications.
Text Books:	
1.	S.Rajasekaran, G.A.Vijayalakshmi Pai, Neural Networks, Fuzzy logic, and genetic algorithms –synthesis and applications-–Prentice-Hall of India private limited, 2008, New Delhi.
Reference Books:	
1.	Timothy J. Ross., Fuzzy Logic with Engineering Applications – Second Edition, Wiley Publications, 2007, New Delhi.
2.	H.J.Zimmerman, Fuzzy set theory and its applications, 4th edition, Springer, 2006, NewDelhi.
3.	S.Nanda and N.R.Das, Fuzzy Mathematical concepts, Narosa Publishing House, New Delhi.
4.	Chennakesava R. Alavala, “Fuzzy Logic and Neural Networks - Basic Concepts & Applications, New Age International (P) Limited, Publishers.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23QTOE01	OE	3	--	--	3	30	70	3 Hrs.

QUANTUM SCIENCE AND TECHNOLOGY

Offered by AIDS, AIML, CIC, CSE, CSBS, IT

(offered to all Branches)

Pre-requisites: Basic Physics, Linear Algebra, and Introduction to Modern Physics.

Course Objectives: Students are expected

1.	To introduce fundamental concepts of quantum mechanics and its mathematical formalism.
2.	To explore quantum computing and communication principles and technologies.
3.	To understand the physical implementation and limitations of quantum systems.
4.	To enable students to relate quantum theory to practical applications in computing, cryptography, and sensing.
5.	To familiarize students with the emerging trends in quantum technologies.

Course Out Comes: At the end of the course students will be able to

S.No	Outcome	Knowledge Level
1.	Apply the Schrodinger equation to solve quantum mechanical problems for systems like the free particle, infinite potential well, and step potential.	K3
2.	Analyze quantum phenomena like superposition and entanglement.	K3
3.	Apply mathematical tools to model and solve quantum systems.	K3
4.	Demonstrate understanding of quantum algorithms and quantum circuits.	K3
5.	Demonstrate the practical use of quantum sensors, materials, and hardware platforms to solve real-world measurement and computing problems.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Fundamentals of Quantum Mechanics:** Historical background, Blackbody radiation, photoelectric effect, and Compton scattering, Dual nature of light and matter, De Broglie hypothesis, Schrodinger equation, Free particle, infinite potential well, step potential, Operators and observables: position, momentum, Hamiltonian, Commutation relations and uncertainty principle, Quantum postulates and measurement theory, Eigenvalues, eigen functions
UNIT-II (10 Hrs)	**Quantum Information Theory:** Classical vs. quantum information, Qubit representation using Bloch sphere, Quantum superposition and quantum entanglement, Dirac notation (bra-ket), tensor products, and composite systems, Bell states and EPR paradox, Quantum gates: Pauli-X, Y, Z; Hadamard; Phase; T; CNOT, Quantum circuit models and notation, Measurement in computational basis, Quantum teleportation and no-cloning theorem, Quantum state tomography (introductory)

UNIT-III (10 Hrs)	**Quantum Computing:** Classical computing review and limitations, Quantum parallelism and interference, Deutsch and Deutsch-Jozsa algorithms, Grover's search algorithm, Oracle and amplitude amplification, Shor's factoring algorithm (overview and significance), Quantum Fourier Transform (QFT), **Quantum error correction:** Bit-flip, phase-flip, and Shor's 9-qubit code, Introduction to quantum programming: Qiskit, Cirq, IBM Quantum Experience (overview)
UNIT-IV (10 Hrs)	**Quantum Communication:** Introduction to quantum cryptography, **Quantum key distribution (QKD):** BB84 protocol, Entanglement-based QKD: Ekert protocol (E91), Eavesdropping and security of QKD, Quantum teleportation (circuit and protocol), Quantum dense coding, Quantum networks and entanglement swapping, Role of quantum repeaters, Single-photon sources and detectors, Implementation challenges (loss, decoherence, noise)
UNIT-V (10 Hrs)	**Quantum Technologies and Applications: Quantum sensors:** magnetometry, gravimetry, **Quantum metrology:** standard time, atomic clocks, Quantum imaging and lithography, **Quantum materials:** topological insulators, graphene, quantum dots, NV centers in diamonds for sensing, **Hardware platforms:** Superconducting qubits, Trapped ions, Photonic quantum processors, Quantum supremacy and NISQ era, Global initiatives: IBM, Google, D-Wave, IonQ, India's NQM, Ethical concerns and future prospects
TEXTBOOK:	
1.	Quantum Computation and Quantum Information by Michael A. Nielsen and Isaac L. Chuang,2nd Edition,**Year:** 2010 (Reprinted 2016/2020),**Publisher:** Cambridge University Press.
2.	Quantum Mechanics: Concepts and Applications by NouredineZettili 2nd Edition,2009 (2nd E Wiley publications.
REFERENCE BOOKS:	
1.	Quantum Computing for Computer Scientists by Noson S. Yanofsky and Mirco A. Mannucci, 1st Edition, 2008, Cambridge University Press.
2.	Benenti G., Casati G. and Strini G., Principles of Quantum Computation and Information, Vol.I: Basic Concepts, Vol II.2nd Combined Edition, ,2018, publisher: World Scientific.
3.	Basic Tools and Special Topics, World Scientific. Pittenger A. O., An Introduction to Quantum Computing Algorithms, 1st Edition,2000 (Reprinted 2012),Birkhäuser Boston (Springer).
E-Resources:	
1.	Core Principles of Quantum Mechanics: https://plato.stanford.edu/entries/qt-quantcomp/ (Stanford Encyclopedia of Philosophy)
2.	Foundations of Quantum Information: https://quantum-computing.ibm.com/composer/docs/iqx/guide/quantum-information (IBM Quantum)
3.	Algorithms and Quantum Computing Models: https://qiskit.org/learn/ (Qiskit (IBM))

4.	Quantum Cryptography and Communication Protocols: https://www.nature.com/articles/nphoton.2014.149 (Nature Photonics)
5.	Emerging Quantum Technologies: https://www.quantum.gov/ (National Quantum Coordination Office (U.S.))